

Modeling, Control and Simulation of Four UAVs Rigidly Connected via a Cross-Shaped Rod

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Abstract—This paper investigates the modeling and control of four rigidly connected quadrotor UAVs (FRCQU) system. A dynamic model based on Euler–Lagrange theory is established, in which the overall system is treated as a single rigid body to accurately capture the coupled translational and rotational dynamics. To achieve stable and coordinated flight, a cascaded PD control architecture is developed for the system. The proposed framework ensures desired motion and robust stability under rigid-body constraints. The simulation results validate its effectiveness in achieving rapid convergence of position errors and coordinated rotor-level response.

Keywords—Rigidly Connected UAVs, Cascaded PD Control, Rotor Speed Distribution

I. INTRODUCTION

Unmanned aerial vehicles (UAVs) have recently developed rapidly, driven by advances in autonomous navigation, sensing and communication technologies. Their versatility supports broad applications including precision agriculture [1], disaster emergency response [2], aerial mapping and cargo delivery [3]. Among these, aerial cargo transportation has drawn significant attention to logistics in remote, congested, or disaster-stricken areas. Previous studies have explored the improvement of load efficiency and system stability through cooperative UAV swarms [3]. However, reliable and efficient multi-UAV load transport remains a challenge due to complex dynamics, communication constraints, and control coordination.

To address the aforementioned challenges, researchers have proposed various modeling and control strategies for multi-UAV cooperative load transportation. In terms of modeling, Liang *et al.* [4] propose a redundant dynamic model for dual quadrotor transport derived from Euler–Lagrange theory, which decouples control inputs from system dynamics, thereby simplifying controller design and stability analysis. Yang *et al.* [5] develop an omni-directional aerial transport system with a fully actuated UAV and dual-cable payload, allowing precise attitude control through dynamic coupling modeling. Jin *et*

al. [6] propose a multi-quadrotor system for transporting a 3D rigid payload, addressing nonlinearities and uncertainties to meet performance and safety constraints. On the control side, an adaptive control framework with a concurrent learning estimator is proposed in [7] for cooperative transport of unknown payloads, allowing robust tracking and online estimation of payload properties. Wang *et al.* [8] presented a robust cooperative controller that combined backstepping and distributed force allocation, allowing dynamic cable re-configuration and precise trajectory tracking. [9] proposes a distributed robust control for UAV swarms using local data and semi-definite programming to ensure stable formation under uncertainties, extending to high-order kinematics. [10] proposes an optimization-based control for multi-UAV payload transport, using Control Barrier Functions to ensure robust obstacle avoidance, satisfy payload constraints, and achieve precise target tracking. These efforts demonstrate the feasibility and robustness of both model-based and control-driven approaches in cooperative UAV payload transport.

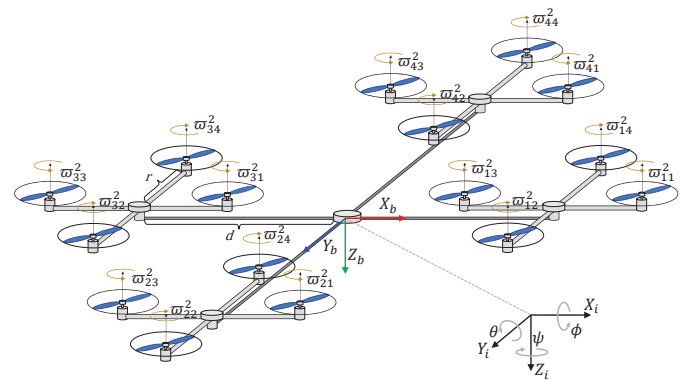


Fig. 1. Configuration of the FRCQU system, consisting of four identical cross-shaped quadrotors rigidly mounted at the ends of a lightweight cross-shaped rod, with each quadrotor's forward direction aligned with the forward direction of the system, that is, the positive X_b axis.

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Although previous studies have achieved notable results, they often rely on complex control models. We propose a novel modeling and control framework for four quadrotor UAVs, which are fixed to the ends of a rigid cross-shaped rod via slot-based mounting mechanisms. This configuration improves the total thrust of the system by increasing the number of UAVs, effectively compensating for the limited payload capacity of a single UAV. Meanwhile, the design of slot-based mounting mechanisms allows each individual UAV to be detached or reconfigured conveniently, offering high scalability. Our main contributions are summarized below.

- 1) We propose a novel Four Rigidly Connected Quadrotor UAVs (FRCQU) system, as shown in Fig. 1. Its overall dynamics are modeled using the Euler–Lagrange formalism, and a rotor thrust–torque mapping framework is established to allocate the rotor speed.
- 2) A cascaded PD control architecture is designed to ensure coordinated motion under the rigid-body constraints of the FRCQU system, where the outer and inner loops are responsible for position and attitude regulation, respectively.
- 3) Comprehensive numerical simulations validate that the proposed FRCQU architecture and cascaded PD controllers achieve robust stability and precise trajectory tracking.

II. PRELIMINARIES AND DYNAMICS MODELING

The configuration of the FRCQU system consists of four identical quadrotor UAVs rigidly affixed to the ends of a lightweight cross-shaped rod, forming an integrated structure. The geometric center of the cross-shaped rod houses a centralized control unit, which synchronously regulates all rotor speeds of four quadrotors, allowing coordinated and unified motion control of the rigid assembly as a whole.

A. Notations

To describe the motion of the FRCQU system, we define two coordinate frames: an inertial frame $\mathcal{F}_i = \{X_i, Y_i, Z_i\}$ and a body-fixed frame $\mathcal{F}_b = \{X_b, Y_b, Z_b\}$ with its origin located at the center of mass of the cross-shaped rod. In the inertial frame $\mathcal{F}_i$, the position of the system is expressed by $\xi = [x \ y \ z]^T \in \mathbb{R}^3$, and its attitude is denoted as Euler angles $\eta = [\phi \ \theta \ \psi]^T \in \mathbb{R}^3$, representing roll, pitch and yaw, respectively. The angular velocity of the system in the body-fixed frame $\mathcal{F}_b$ is defined as $\omega = [p \ q \ r]^T$. The rotation matrix that maps vectors from $\mathcal{F}_b$ to $\mathcal{F}_i$ based on the XYZ Euler angle sequence is presented in [11]:

$$R(\eta) = \begin{bmatrix} c_\psi c_\theta & c_\psi s_\theta s_\phi - s_\psi c_\phi & c_\psi s_\theta c_\phi + s_\psi s_\phi \\ s_\psi c_\theta & s_\psi s_\theta s_\phi + c_\psi c_\phi & s_\psi s_\theta c_\phi - c_\psi s_\phi \\ -s_\theta & c_\theta s_\phi & c_\theta c_\phi \end{bmatrix} \quad (1)$$

where $c(\cdot)$ and $s(\cdot)$ represent $\cos(\cdot)$ and $\sin(\cdot)$, respectively.

B. Dynamics Modeling of the FRCQU System

In this section, we model the dynamics of the FRCQU system based on the Euler–Lagrange method. Suppose that

the ground is the reference point for the zero potential, then the gravitational potential energy is: $U = -mgz$. Due to the symmetry of the system, the inertia matrix is simplified as $I = \text{diag}(I_{xx}, I_{yy}, I_{zz})$. The translational and rotational kinetic energies of the system are expressed as (see Appendix A for detailed derivations):

$$T_{\text{trans}} = \frac{1}{2} m \dot{\xi}^T \dot{\xi} \quad (2)$$

$$T_{\text{rot}} = \frac{1}{2} \dot{\eta}^T \mathbb{J}(\eta) \dot{\eta} \quad (3)$$

where the generalized inertia matrix $\mathbb{J}(\eta)$ is explicitly given as:

$$\mathbb{J} = \begin{bmatrix} I_{xx} & 0 & -I_{xx} s_\theta \\ 0 & I_{yy} c_\phi^2 + I_{zz} s_\phi^2 & (I_{yy} - I_{zz}) c_\phi s_\phi c_\theta \\ -I_{xx} s_\theta & (I_{yy} - I_{zz}) c_\phi s_\phi c_\theta & I_{xx} s_\theta^2 + I_{yy} s_\phi^2 c_\theta^2 + I_{zz} c_\phi^2 c_\theta^2 \end{bmatrix} \quad (4)$$

Then, the complete motion equations of the FRCQU system are derived using the Euler–Lagrange formalism. The generalized coordinates of a rigid body that is undergoing motion in a three-dimensional space are given by $q = [x \ y \ z \ \phi \ \theta \ \psi]^T \in \mathbb{R}^6$. Based on (2) and (3), the Lagrangian function is defined as follows:

$$L(q, \dot{q}) = T_{\text{trans}} + T_{\text{rot}} - U \quad (5)$$

According to [12], the Euler–Lagrange equation with generalized forces is:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = \begin{bmatrix} f_\xi \\ \tau \end{bmatrix} \quad (6)$$

where $f_\xi = f R(\eta) e_3$, with $e_3 = [0 \ 0 \ 1]^T$ denoting the unit vector of the Z -axis in $\mathcal{F}_b$, $f \in \mathbb{R}$ is the total thrust and $\tau = [\tau_x \ \tau_y \ \tau_z]^T$ is the torque vector.

The translational movement derived from (6) can be expressed by the following equation:

$$m \ddot{\xi} = -f_\xi + m g e_3 \quad (7)$$

where (7) can be expanded into its components as:

$$\begin{cases} \ddot{x} = -\frac{f}{m} (\cos \psi \sin \theta \cos \phi + \sin \psi \sin \phi) \\ \ddot{y} = -\frac{f}{m} (\sin \psi \sin \theta \cos \phi - \cos \psi \sin \phi) \\ \ddot{z} = g - \frac{f}{m} \cos \theta \cos \phi \end{cases} \quad (8)$$

The rotational motion governed by (6) for angular coordinates η can be written as:

$$\mathbb{J}(\eta) \ddot{\eta} + C(\eta, \dot{\eta}) \dot{\eta} = \tau \quad (9)$$

where $C(\eta, \dot{\eta}) \in \mathbb{R}^{3 \times 3}$ represents the Coriolis matrix, which accounts for gyroscopic coupling and centrifugal forces in the rotational dynamics of the system and given by:

$$C(\eta, \dot{\eta}) = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix} \quad (10)$$

where individual elements c_{ij} are:

$$\begin{cases} c_{11} = 0 \\ c_{12} = (I_{yy} - I_{zz})(\dot{\theta}c_\phi s_\phi + \dot{\psi}s_\phi^2 c_\theta - \dot{\psi}c_\phi^2 c_\theta) - I_{xx}\dot{\psi}c_\theta \\ c_{13} = (I_{zz} - I_{yy})\dot{\psi}c_\phi s_\phi c_\theta^2 \\ c_{21} = (I_{zz} - I_{yy})(\dot{\theta}c_\phi s_\phi + \dot{\psi}s_\phi^2 c_\theta - \dot{\psi}c_\phi^2 c_\theta) + I_{xx}\dot{\psi}c_\theta \\ c_{22} = (I_{zz} - I_{yy})\dot{\phi}c_\phi s_\phi \\ c_{23} = -I_{xx}\dot{\psi}s_\theta c_\theta + I_{yy}\dot{\psi}s_\phi^2 c_\theta s_\theta + I_{zz}\dot{\psi}c_\phi^2 c_\theta s_\theta \\ c_{31} = (I_{yy} - I_{zz})\dot{\psi}c_\theta^2 s_\phi c_\phi - I_{xx}\dot{\theta}c_\theta \\ c_{32} = (I_{zz} - I_{yy})(\dot{\theta}c_\phi s_\phi s_\theta + \dot{\phi}s_\phi^2 c_\theta - \dot{\phi}c_\phi^2 c_\theta) \\ \quad + I_{xx}\dot{\psi}s_\theta c_\theta - I_{yy}\dot{\psi}s_\phi^2 c_\theta s_\theta - I_{zz}\dot{\psi}c_\phi^2 c_\theta s_\theta \\ c_{33} = (I_{yy} - I_{zz})\dot{\phi}c_\phi s_\phi c_\theta^2 - I_{yy}\dot{\theta}s_\phi^2 c_\theta s_\theta \\ \quad - I_{zz}\dot{\theta}c_\phi^2 c_\theta s_\theta + I_{xx}\dot{\theta}c_\theta s_\theta \end{cases} \quad (11)$$

Thus, the rotational dynamics of the FRCQU system, obtained via the Euler-Lagrange method, can be formulated as:

$$\ddot{\eta} = \mathbb{J}^{-1}(\tau - C(\eta, \dot{\eta})\dot{\eta}) \quad (12)$$

Therefore, the overall dynamics are described by (8) and (12).

III. THRUST AND TORQUE CALCULATION

For the considered FRCQU system, let ϖ_{ij} denote the rotational speed of the j -th rotor on the i -th quadrotor, where $i = 1, 2, 3, 4$ and $j = 1, 2, 3, 4$. Then, the corresponding thrust force $f_{ij} \in \mathbb{R}$ and torque $\tau_{ij} \in \mathbb{R}$ generated by each rotor are given by:

$$f_{ij} = b\varpi_{ij}^2, \quad \tau_{ij} = \kappa\varpi_{ij}^2 \quad (13)$$

where $b \in \mathbb{R}$ and $\kappa \in \mathbb{R}$ are aerodynamic coefficients dependent on factors such as air density, propeller radius, and the blade's geometric [13].

A. Total System Thrust Modeling

The total thrust of quadrotor i is given by summing the thrust of all its four rotors:

$$f_i = \sum_{j=1}^4 f_{ij} = b \sum_{j=1}^4 \varpi_{ij}^2 \quad (14)$$

Consequently, the overall system thrust becomes:

$$f = \sum_{i=1}^4 f_i = b \sum_{i=1}^4 \sum_{j=1}^4 \varpi_{ij}^2 \quad (15)$$

B. Total System Torque Modeling

The torques generated by each quadrotor are given below.

Quadrotor 1:

$$\begin{cases} \tau_{x1} = rb(\varpi_{14}^2 - \varpi_{12}^2) \\ \tau_{y1} = db(\varpi_{12}^2 + \varpi_{14}^2) + (d-r)b\varpi_{13}^2 + (d+r)b\varpi_{11}^2 \\ \tau_{z1} = \kappa(\varpi_{11}^2 - \varpi_{12}^2 + \varpi_{13}^2 - \varpi_{14}^2) \end{cases} \quad (16)$$

Quadrotor 2:

$$\begin{cases} \tau_{x2} = -db(\varpi_{21}^2 + \varpi_{23}^2) - (d-r)b\varpi_{24}^2 - (d+r)b\varpi_{22}^2 \\ \tau_{y2} = rb(\varpi_{21}^2 - \varpi_{23}^2) \\ \tau_{z2} = \kappa(\varpi_{21}^2 - \varpi_{22}^2 + \varpi_{23}^2 - \varpi_{24}^2) \end{cases} \quad (17)$$

Quadrotor 3:

$$\begin{cases} \tau_{x3} = rb(\varpi_{34}^2 - \varpi_{32}^2) \\ \tau_{y3} = -db(\varpi_{32}^2 + \varpi_{34}^2) - (d+r)b\varpi_{33}^2 - (d-r)b\varpi_{31}^2 \\ \tau_{z3} = \kappa(\varpi_{31}^2 - \varpi_{32}^2 + \varpi_{33}^2 - \varpi_{34}^2) \end{cases} \quad (18)$$

Quadrotor 4:

$$\begin{cases} \tau_{x4} = db(\varpi_{41}^2 + \varpi_{43}^2) + (d-r)b\varpi_{42}^2 + (d+r)b\varpi_{44}^2 \\ \tau_{y4} = rb(\varpi_{41}^2 - \varpi_{43}^2) \\ \tau_{z4} = \kappa(\varpi_{41}^2 - \varpi_{42}^2 + \varpi_{43}^2 - \varpi_{44}^2) \end{cases} \quad (19)$$

where $r \in \mathbb{R}$ represents the distance from the center of the rotor to the center of the UAV, and $d \in \mathbb{R}$ denotes the distance between the center of the UAV and the center of the cross-shaped rod.

The total torque of the system is obtained by summing the torque components given in (16), (17), (18) and (19):

$$\tau_x = \sum_{i=1}^4 \tau_{xi}, \quad \tau_y = \sum_{i=1}^4 \tau_{yi}, \quad \tau_z = \sum_{i=1}^4 \tau_{zi} \quad (20)$$

C. Minimum Rotor Speed Allocation

Based on (15) and (20), the total thrust f and the torque vector τ of the system can be represented by:

$$y = \begin{bmatrix} f \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = M\varpi \quad (21)$$

where

$$M^T = \begin{bmatrix} b & 0 & (d+r)b & \kappa \\ b & -rb & db & -\kappa \\ b & 0 & (d-r)b & \kappa \\ b & rb & db & -\kappa \\ b & -db & rb & \kappa \\ b & -(d+r)b & 0 & -\kappa \\ b & -db & -rb & \kappa \\ b & -(d-r)b & 0 & -\kappa \\ b & 0 & -(d-r)b & \kappa \\ b & -rb & -db & -\kappa \\ b & 0 & -(d+r)b & \kappa \\ b & rb & -db & -\kappa \\ b & db & rb & \kappa \\ b & (d-r)b & 0 & -\kappa \\ b & db & -rb & \kappa \\ b & (d+r)b & 0 & -\kappa \end{bmatrix}, \quad \varpi = \begin{bmatrix} \varpi_{11}^2 \\ \varpi_{12}^2 \\ \varpi_{13}^2 \\ \varpi_{14}^2 \\ \varpi_{21}^2 \\ \varpi_{22}^2 \\ \varpi_{23}^2 \\ \varpi_{24}^2 \\ \varpi_{31}^2 \\ \varpi_{32}^2 \\ \varpi_{33}^2 \\ \varpi_{34}^2 \\ \varpi_{41}^2 \\ \varpi_{42}^2 \\ \varpi_{43}^2 \\ \varpi_{44}^2 \end{bmatrix} \quad (22)$$

Since the matrix $M \in \mathbb{R}^{4 \times 16}$ is full rank in rows but not square, the system $M\varpi = y$ is underdetermined and admits

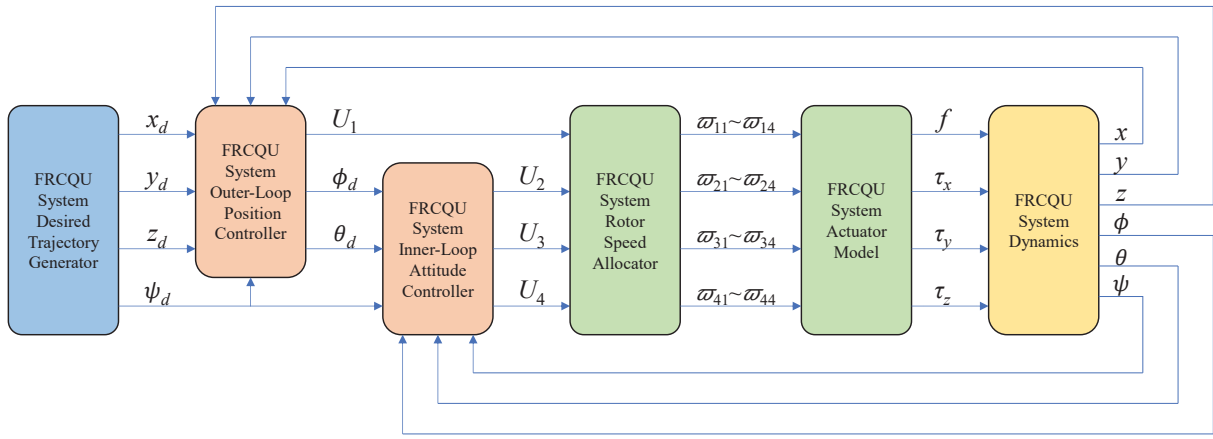


Fig. 2. Control architecture for the FRCQU system.

infinitely many solutions. To select a unique solution, we aim to find one with the minimum Euclidean norm, i.e., the solution subject to a constraint that minimizes $\|\varpi\|_2$. Hence, the minimum-norm solution is:

$$\varpi_{\min} = M^T(MM^T)^{-1}y \quad (23)$$

This solution ensures that, among all possible solutions that satisfy the system, the one with the smallest energy is chosen.

IV. CONTROLLER DESIGN

From a dynamics perspective, the FRCQU system, with four inputs and six outputs, is inherently unstable. Therefore, designing a control strategy that can stabilize such a system presents significant challenges. This paper proposes an inner–outer loop PD control architecture shown in Fig. 2.

The cascaded controller employs a two-layer structure, where position tracking is handled by the outer loop, and attitude regulation is assigned to the inner loop. Specifically, the outer loop receives the desired position $\xi_d = [x_d \ y_d \ z_d]^T$ and the desired yaw angle ψ_d . Then it computes a translational acceleration control signal $u_\xi = [u_x \ u_y \ u_z]^T$, which is subsequently used to determine the total thrust U_1 , and the desired roll and pitch angles ϕ_d and θ_d . These quantities are derived from the translational dynamics described in (8):

$$U_1 = m\sqrt{u_x^2 + u_y^2 + (u_z - g)^2} \quad (24)$$

$$\phi_d = \arcsin\left(\frac{u_y \cos(\psi_d) - u_x \sin(\psi_d)}{U_1/m}\right) \quad (25)$$

$$\theta_d = \arcsin\left(\frac{u_x \cos(\psi_d) + u_y \sin(\psi_d)}{u_z - g}\right) \quad (26)$$

The inner loop regulates the attitude of the system according to the desired angles provided by the outer loop. Specifically, its input is the desired attitude angles generated by the outer loop, expressed as $\eta_d = [\phi_d \ \theta_d \ \psi_d]^T$. The output of the inner loop is a virtual angular acceleration control signal, given by $u_\eta = [u_\phi \ u_\theta \ u_\psi]^T$.

A. Position Control Design

Define the position tracking error as $x_e = x_d - x$, with its time derivative given by $\dot{x}_e = \dot{x}_d - \dot{x}$. The position PD control law in the x direction is formulated as follows:

$$\ddot{x} = u_x = k_{px}x_e + k_{dx}\dot{x}_e + \ddot{x}_d \quad (27)$$

It then follows from (27) that:

$$\ddot{x}_e + k_{dx}\dot{x}_e + k_{px}x_e = 0 \quad (28)$$

where $\ddot{x}_e = \ddot{x}_d - \ddot{x}$.

To ensure stability in the x direction, the characteristic equation

$$s^2 + k_{dx}s + k_{px} = 0 \quad (29)$$

must have all its roots located in the left-half complex plane. This condition is satisfied if $k_{px} > 0$ and $k_{dx} > 0$, where k_{px} and k_{dx} are the proportional coefficient and derivative coefficient, respectively.

Similarly, the position PD control laws in the y and z directions are given by:

$$\ddot{y} = u_y = k_{py}y_e + k_{dy}\dot{y}_e + \ddot{y}_d \quad (30)$$

$$\ddot{z} = u_z = k_{pz}z_e + k_{dz}\dot{z}_e + \ddot{z}_d \quad (31)$$

where the control gains must satisfy $k_{py}, k_{dy}, k_{pz}, k_{dz} > 0$, ensuring that all eigenvalues of the system lie in the left-half of the complex plane.

B. Attitude Control Design

Define the tracking error as $\eta_e = \eta_d - \eta$ and its time derivative as $\dot{\eta}_e = \dot{\eta}_d - \dot{\eta}$. The PD control laws for attitude tracking are given by:

$$\ddot{\phi} = u_\phi = k_{p\phi}\phi_e + k_{d\phi}\dot{\phi}_e + \ddot{\phi}_d \quad (32)$$

$$\ddot{\theta} = u_\theta = k_{p\theta}\theta_e + k_{d\theta}\dot{\theta}_e + \ddot{\theta}_d \quad (33)$$

$$\ddot{\psi} = u_\psi = k_{p\psi}\psi_e + k_{d\psi}\dot{\psi}_e + \ddot{\psi}_d \quad (34)$$

where the controller parameters $k_{p\phi}, k_{d\phi}, k_{p\theta}, k_{d\theta}, k_{p\psi}, k_{d\psi}$ must be positive to ensure convergence and stability.

Since the attitude dynamics has a second-order structure similar to that of the position dynamics, the stability analysis follows the same principle and is omitted here for brevity.

C. Desired Rotor Speeds

Based on (9), (32), (33) and (34), the control torques U_2 , U_3 , and U_4 corresponding to roll, pitch, and yaw, respectively, can be computed as:

$$U_2 = \mathbb{J}(\eta)u_\phi + C(\eta, \dot{\eta})\dot{\phi} \quad (35)$$

$$U_3 = \mathbb{J}(\eta)u_\theta + C(\eta, \dot{\eta})\dot{\theta} \quad (36)$$

$$U_4 = \mathbb{J}(\eta)u_\psi + C(\eta, \dot{\eta})\dot{\psi} \quad (37)$$

Together with the total thrust U_1 from (24), the complete control input vector is given by $U = [U_1 \ U_2 \ U_3 \ U_4]^T$. To determine the desired squared rotor speeds $\varpi_d \in \mathbb{R}^{16}$, a minimum norm solution is computed from (23) as follows:

$$\varpi_d = M^T(MM^T)^{-1}U \quad (38)$$

V. SIMULATION RESULTS AND ANALYSIS

All simulations were conducted using *Simulink* R2024b, where the dynamic model and controller were implemented using the built-in fixed-step numerical solver `ode4` (Runge-Kutta) with a step size of 0.001 s. The simulation of the FRCQU system was carried out using the following model parameters: $m = 5.6$ kg, $g = 9.81$ m/s², $r = 0.225$ m, $d = 0.5$ m, $J_{xx} = J_{yy} = 0.2306$ kg·m², $J_{zz} = 0.4087$ kg·m², $b = 1.105 \times 10^{-5}$ N·s², $\kappa = 1.779 \times 10^{-7}$ N·m·s². The initial position of the system is $(x, y, z) = (0, 0, 0)$ m, and the gains in PD control are listed in Table I.

TABLE I
PD CONTROL GAINS

Variable	x	y	z	ϕ	θ	ψ
k_{pi}	81/16	81/16	9/4	36	36	25/9
k_{di}	9/2	9/2	3	12	12	10/3

The FRCQU system executes a structured mission consisting of four sequential phases, where the desired yaw angle is set to $\psi_d = 0^\circ$ throughout the trajectory and the desired positions are expressed in the inertial frame $\mathcal{F}_i$ as (x_d, y_d, z_d) .

- 1) **Vertical Ascent:** The system takes off from the origin at $(0, 0, 0)$ m and ascends along the $-Z_i$ axis to reach $(0, 0, -1.5)$ m.
- 2) **Horizontal Translation:** From position $(0, 0, -1.5)$ m, the system moves along the $+X_i$ direction until it reaches $(1, 0, -1.5)$ m.
- 3) **Circular Tracking:** Starting at $(1, 0, -1.5)$ m, the system follows a clockwise circular path in the X_i - Y_i plane with a radius of 1 m, i.e., $x^2 + y^2 = 1$. After completing two loops and passing through the point

$(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, -1.5)$ m for the last time, it returns to $(1, 0, -1.5)$ m.

- 4) **Descent and Landing:** From $(1, 0, -1.5)$ m, the UAV descends vertically along the $+Z_i$ axis and lands at the origin $(0, 0, 0)$ m.

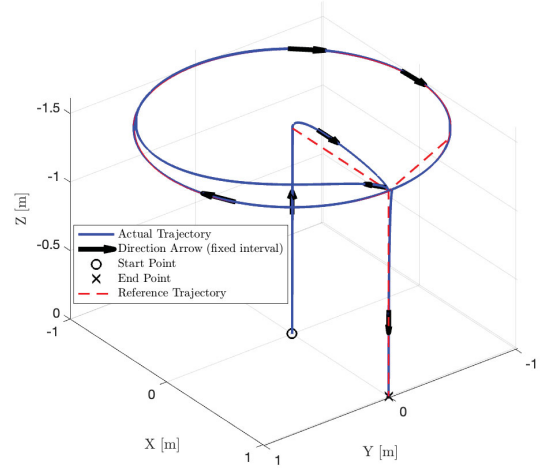


Fig. 3. 3D Position Tracking of the FRCQU System.

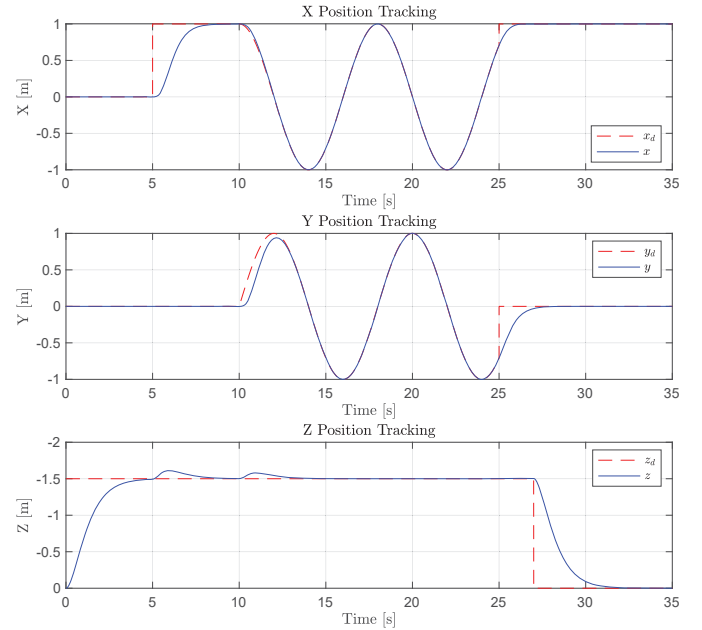


Fig. 4. Desired Position Tracking of the FRCQU System.

The overall 3D trajectory of the system is illustrated in Fig. 3, with detailed tracking performance along the X_i , Y_i and Z_i axes depicted in Fig. 4. The simulation results demonstrate that the presented modeling and control strategy for the rigidly connected quadrotor cluster enables effective position convergence and coordinated rotor-level adaptation, as shown in Fig. 4 and Fig. 5. These results validate the effectiveness of the formulated framework in accomplishing both translational and rotational coordination under disturbance-free conditions.

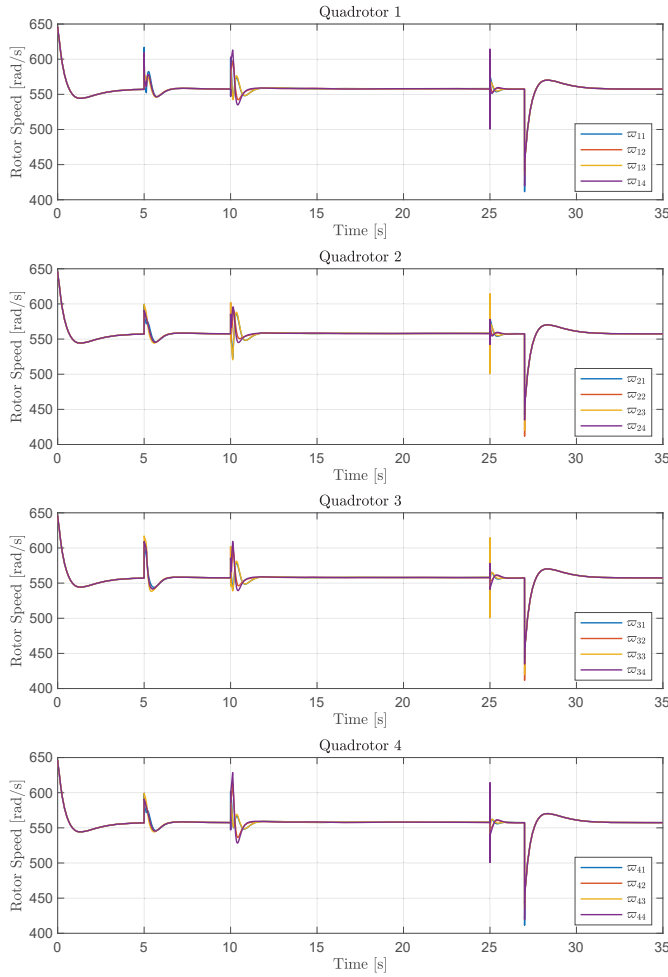


Fig. 5. Rotor Speeds of the FRCQU system.

VI. CONCLUSION

This paper presents a comprehensive dynamic model of the system composed of four rigidly connected quadrotor UAVs using the Euler–Lagrange approach. Based on the proposed model, a cascaded PD control scheme is developed to achieve stable attitude regulation and precise position tracking under rigid-body constraints. Owing to slot-based mounting mechanisms, each UAV can be easily assembled or disassembled, providing the system with high scalability. For example, depending on the weight of the cargo, users can choose to deploy a single UAV or the complete FRCQU system to meet the required payload capacity. This construction shows significant potential for applications in cooperative aerial load transportation, particularly in scenarios requiring increased payload capacity and coordinated maneuvering.

Although the proposed framework shows promising performance in simulation, this study is limited to numerical validation due to resource and platform constraints. Real-world factors such as aerodynamic disturbances, sensor noise, and actuator delays are not yet considered, which may affect performance in practical scenarios. In future work, we will

focus on physical experiments to evaluate robustness and refine the controller under real-world conditions.

APPENDIX

A. Derivation of the Rotational Kinetic Energy

To express the rotational kinetic energy via the Euler angle rates, we relate the angular velocity ω to the Euler angle derivative $\dot{\eta}$ via the Jacobian matrix W_η :

$$\dot{\eta} = W_\eta^{-1} \omega \quad (39)$$

where W_η is given by:

$$W_\eta = \begin{bmatrix} 1 & 0 & -s_\theta \\ 0 & c_\phi & s_\phi c_\theta \\ 0 & -s_\phi & c_\phi c_\theta \end{bmatrix} \quad (40)$$

Substituting (40) into the standard expression of rotational kinetic energy:

$$T_{\text{rot}} = \frac{1}{2} \omega^T I \omega = \frac{1}{2} \dot{\eta}^T (W_\eta^T I W_\eta) \dot{\eta} \quad (41)$$

Therefore, the generalized inertia matrix can be defined as:

$$\mathbb{J}(\eta) = W_\eta^T I W_\eta \quad (42)$$

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